HW4

Alex Villagomez

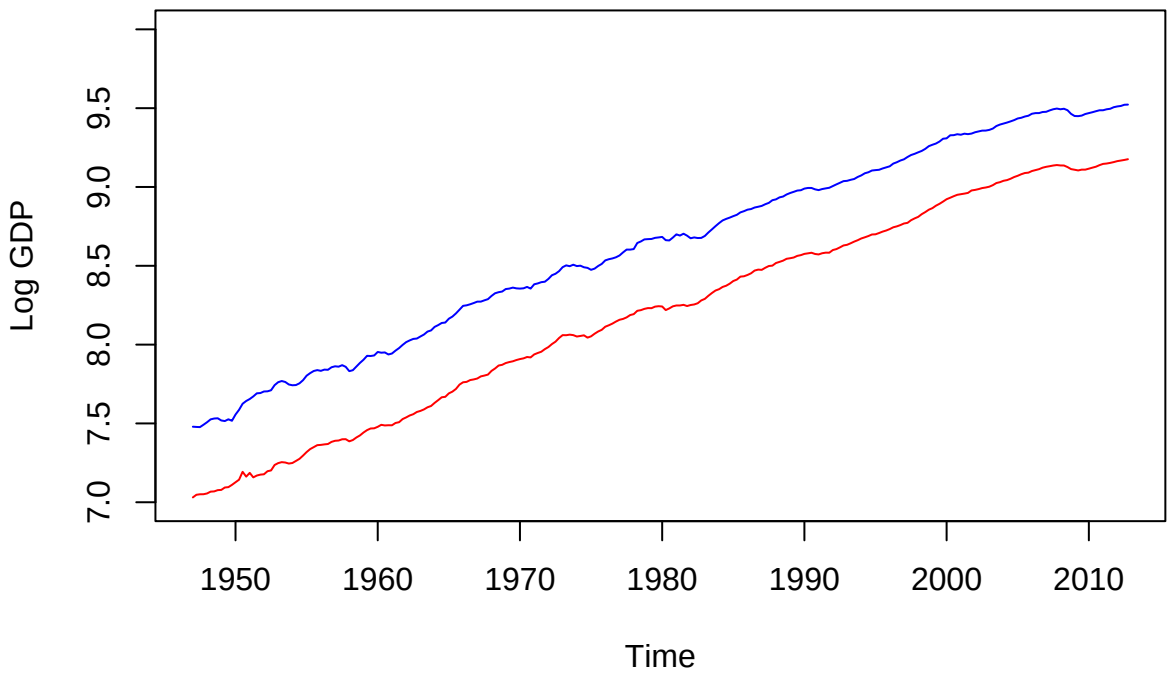
2026-03-01

Contents

Problem 1

```
data=read_excel("../../data/Chapter12_exercises_data.xls", sheet="Exercise 1-2")
y=as.numeric(data$GDP, na.omit=TRUE)
c=as.numeric(data$CONSUMPTION, na.omit=TRUE)
y=ts(y, start=c(1947,1), end=c(2012,4), freq=4)
c=ts(c, start=c(1947,1), end=c(2012,4), freq=4)
ly=log(y)
lc=log(c)

plot(ly, type="l", col="blue", ylab="Log GDP", xlab="Time", ylim=c(7, 10))
lines(lc, col="red")
```



```r
adf.test(ly)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ly
## Dickey-Fuller = -1.0655, Lag order = 6, p-value = 0.9266
## alternative hypothesis: stationary
```

```r
adf.test(lc)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  lc
## Dickey-Fuller = -0.48699, Lag order = 6, p-value = 0.982
## alternative hypothesis: stationary
```

```r
adf.test(diff(ly))
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  diff(ly)
## Dickey-Fuller = -6.4239, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

```r
adf.test(diff(lc))
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  diff(lc)
## Dickey-Fuller = -5.6799, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

The two time series are non-stationary which is a good requirement for cointegration. They are also economically correlated and share similar dynamics in the plot. Their differences are stationary, which is a good requirement for cointegration.

```r
comodel=lm(lc~ly)
summary(comodel)
```

```
##
## Call:
## lm(formula = lc ~ ly)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.043327 -0.014312 -0.003026  0.012879  0.086937
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.961590   0.018569  -51.78   <2e-16 ***
## ly           1.062277   0.002149  494.31   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0215 on 262 degrees of freedom
## Multiple R-squared:  0.9989, Adjusted R-squared:  0.9989
## F-statistic: 2.443e+05 on 1 and 262 DF,  p-value: < 2.2e-16
```

```
resid=residuals(comodel)
adf.test(resid)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  resid
## Dickey-Fuller = -3.4222, Lag order = 6, p-value = 0.05122
## alternative hypothesis: stationary
```

After running the model, we can see that the R^2 is very high which is good for cointegration. However, the residuals are not stationary because the p-value (.051) is above 5%. Therefore, this is not a cointegrated relationship at the 5% level, although the result is borderline.

Problem 2

```
ci=cbind(ly,lc)
jt=ca.jo(ci)
summary(jt)
```

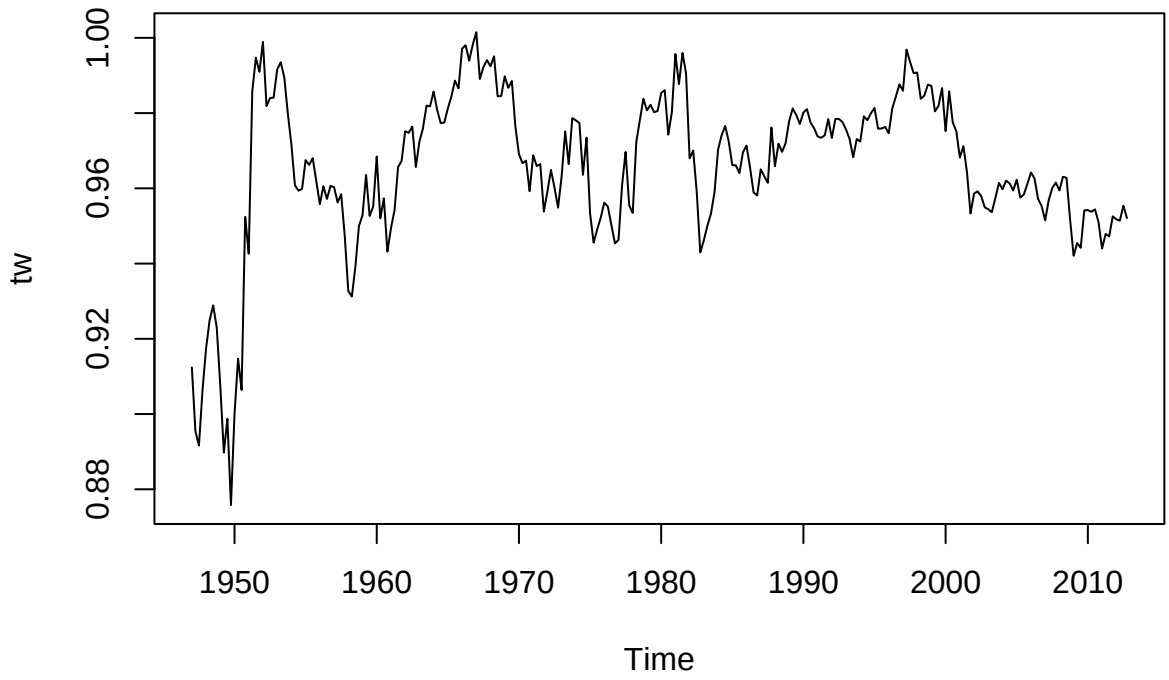
```
##
## ######################
## # Johansen-Procedure #
## ######################
##
## Test type: maximal eigenvalue statistic (lambda max) , with linear trend
##
## Eigenvalues (lambda):
## [1] 0.06557820 0.01278853
##
## Values of teststatistic and critical values of test:
##
##           test 10pct  5pct  1pct
## r <= 1 |  3.37  6.50  8.18 11.65
## r = 0  | 17.77 12.91 14.90 19.19
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##            ly.l2     lc.l2
## ly.l2  1.0000000  1.000000
```

```
## lc.l2 -0.9340406 -1.087854
##
## Weights W:
## (This is the loading matrix)
##
##             ly.l2       lc.l2
## ly.d -0.087449481 0.006631659
## lc.d  0.005178539 0.009423388
```

```
tw=ly-.934*lc
adf.test(tw)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  tw
## Dickey-Fuller = -3.3975, Lag order = 6, p-value = 0.05535
## alternative hypothesis: stationary
```

```
plot(tw)
```



The Johansen maximal eigenvalue test rejects r = 0 at the 5% level (17.77 > 14.90) but not at the 1% level, and does not reject r <= 1, which points to one cointegrating relationship. However, the adf test on the combination given by the first eigenvector (ly - .934lc) has a p-value of .055, so it is not stationary at the 5% level. Together with Problem 1, the evidence for a cointegrating relationship between GDP and consumption is borderline.

Problem 3

```
data=read_excel("../../data/Chapter12_exercises_data.xls", sheet="Exercise 3")
cpigermany=ts(data$CPI_Germany, start=c(1999,1), end=c(2013,1), freq=12)
cpius=ts(data$CPI_US, start=c(1999,1), end=c(2013,1), freq=12)
ex=ts(data$EX_dollar_euro, start=c(1999,1), end=c(2013,1), freq=12)

wt=cpius-ex*cpigermany
adf.test(wt)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  wt
## Dickey-Fuller = -2.5342, Lag order = 5, p-value = 0.3533
## alternative hypothesis: stationary
```

After running the adf test on the residuals of cpius-ex*cpigermany, we can see that the residuals are not stationary because the p-value is 35%. This means that there is not a cointegrated relationship and that the PPP does not hold.

Problem 4

```
data=read_excel("../../data/Chapter12_exercises_data.xls", sheet="Exercise 7")
la=ts(data$ILA, start=c(1975,1), end=c(2012,4), freq=4)
rv=ts(data$IRV, start=c(1975,1), end=c(2012,4), freq=4)

ci=cbind(la,rv)
jt=ca.jo(ci)
summary(jt)
```

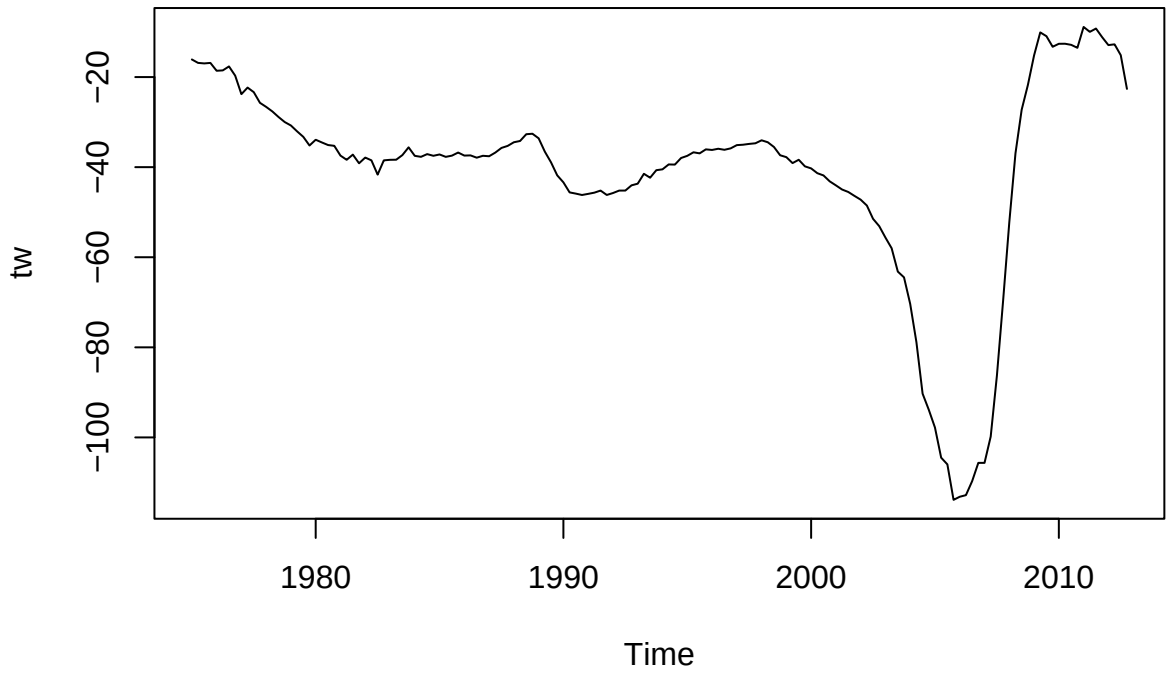
```
##
## ######################
## # Johansen-Procedure #
## ######################
##
## Test type: maximal eigenvalue statistic (lambda max) , with linear trend
##
## Eigenvalues (lambda):
## [1] 0.195512338 0.000101253
##
## Values of teststatistic and critical values of test:
##
##           test 10pct  5pct  1pct
## r <= 1 |  0.02  6.50  8.18 11.65
## r = 0  | 32.63 12.91 14.90 19.19
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##           la.l2      rv.l2
## la.l2  1.000000  1.0000000
## rv.l2 -1.431533 -0.8885865
```

```
##
## Weights W:
## (This is the loading matrix)
##
##           la.l2        rv.l2
## la.d 0.01284897 0.0013312024
## rv.d 0.03739277 0.0009409022
```

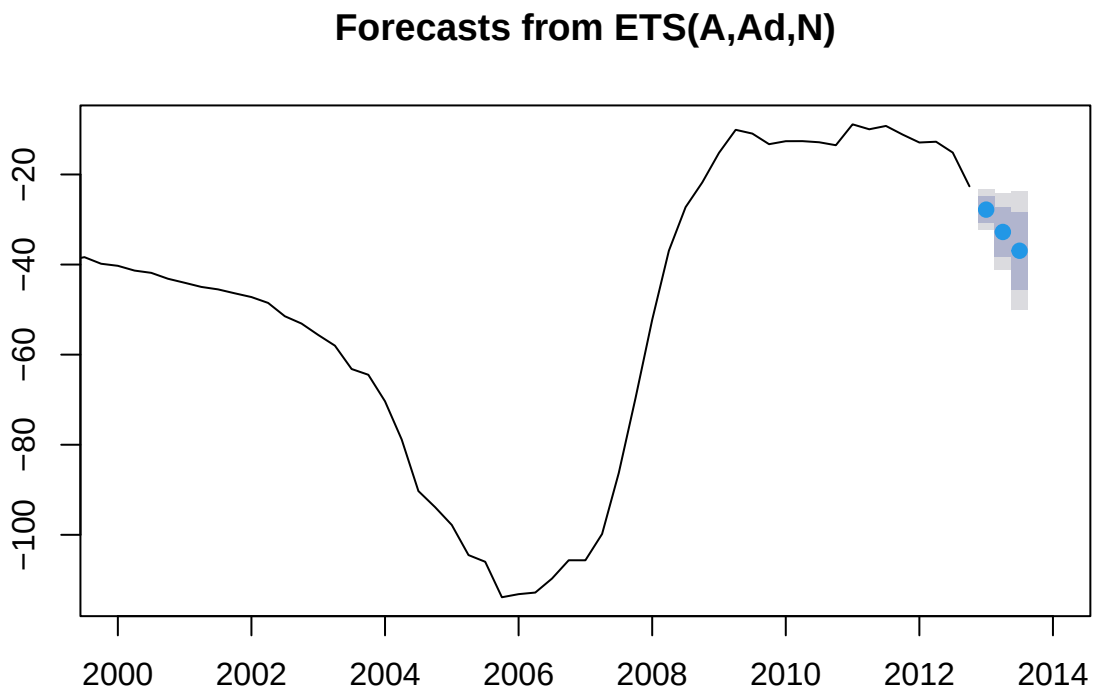
```
tw=la-1.4315*rv
adf.test(tw)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  tw
## Dickey-Fuller = -3.8471, Lag order = 5, p-value = 0.01861
## alternative hypothesis: stationary
```

```
plot(tw)
```




```
plot(forecast(ets(tw), h=3), xlim=c(2000,2014))
```



Yes, from the adf test on the johansen test eigenvectors, we can see that they are stationary which means that there is a cointegrated relationship. We can see that the forecast shows that the series is going to mean revert to the long run equilibrium.

Problem 5

```
set.seed(144)
spec=ugarchspec(
variance.model = list(model = "sGARCH", garchOrder = c(4, 0)),
mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
distribution.model = "norm")
setfixed(spec) <- list(
  mu = 0,
  omega = 0.1,
  alpha1 = 0.20,
  alpha2 = 0.15,
  alpha3 = 0.10,
  alpha4 = 0.05
)

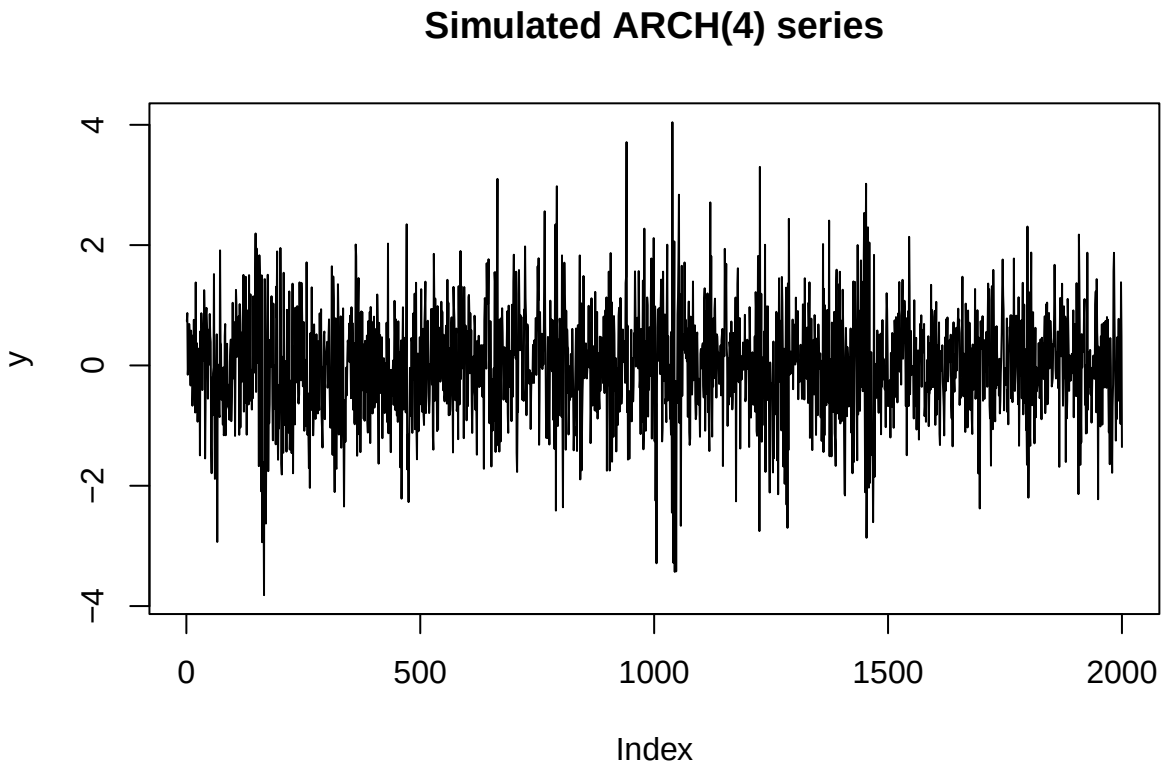
sim <- ugarchpath(spec, n.sim = 2000, n.start = 500, m.sim = 1)

mu  <- as.numeric(fitted(sim))
eps <- as.numeric(sim@path$seriesSim)
y   <- mu + eps

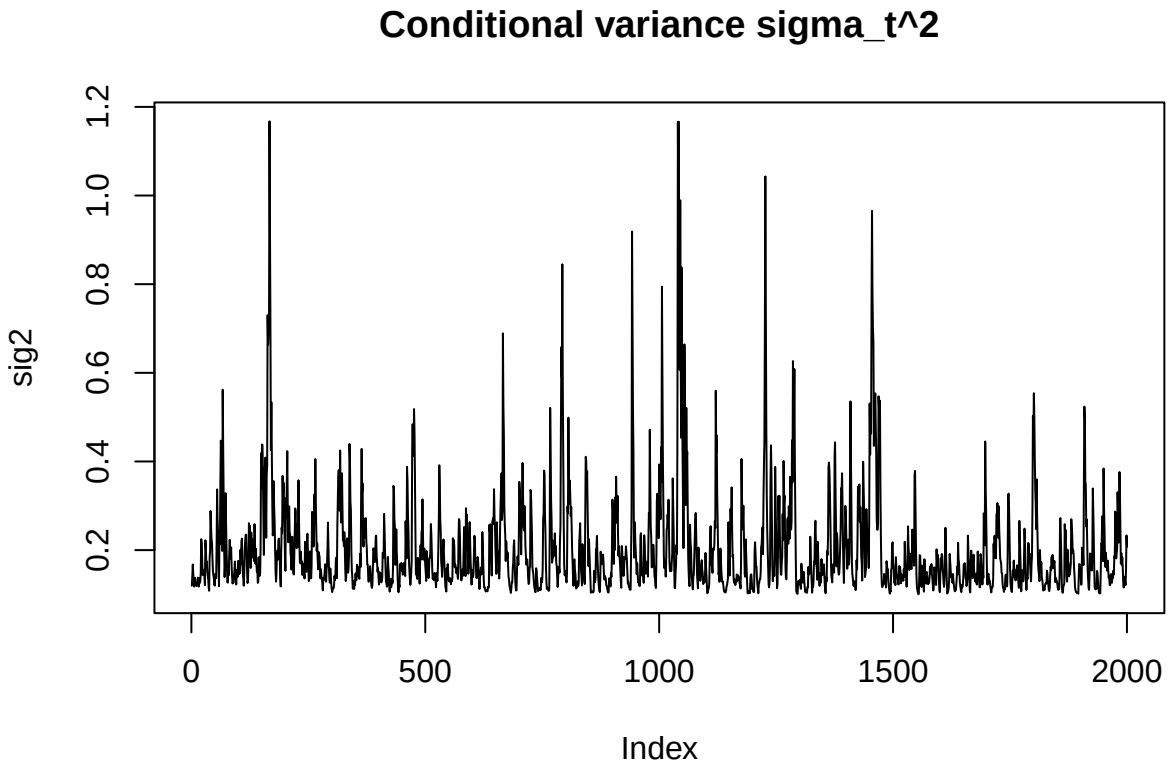
sig  <- as.numeric(sigma(sim))
```

```r
sig2 <- sig^2
z    <- eps / sig

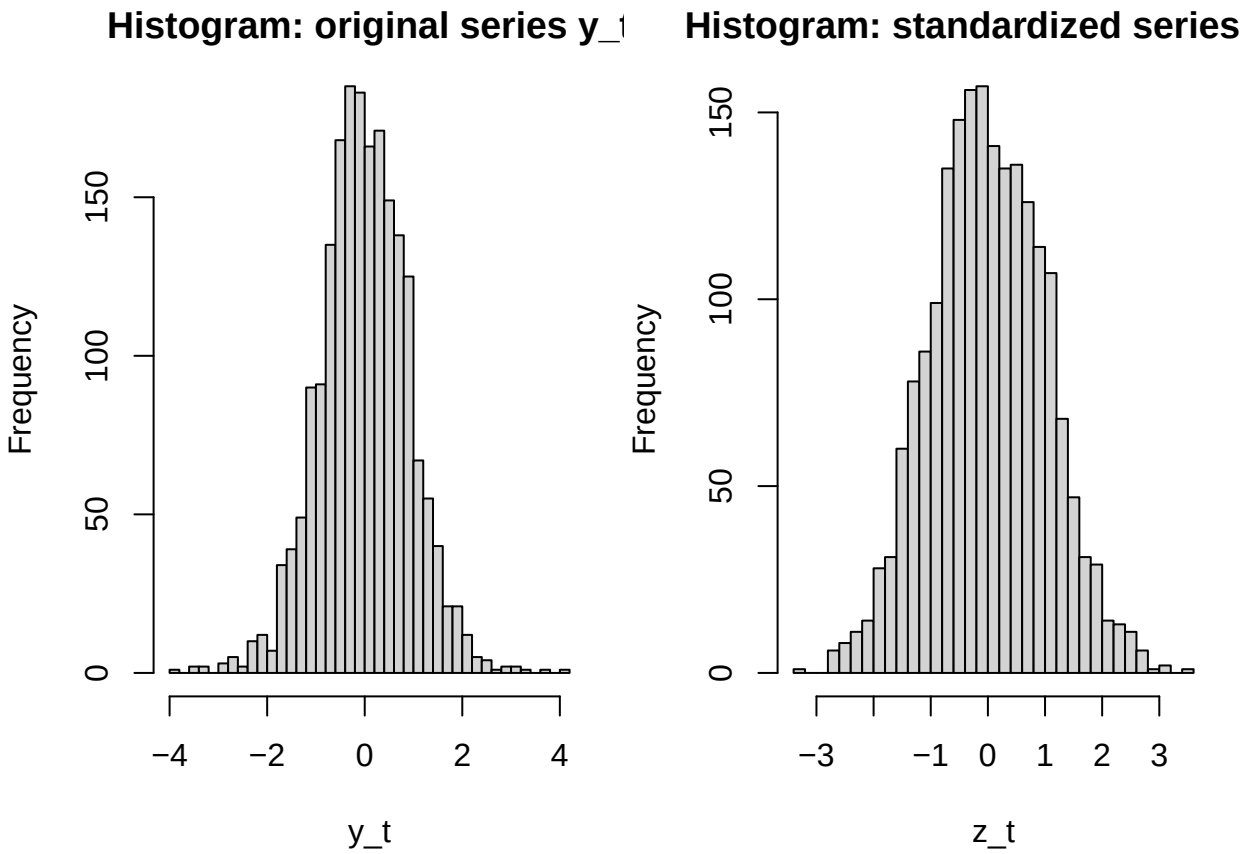
plot(y, type="l", main="Simulated ARCH(4) series")
```



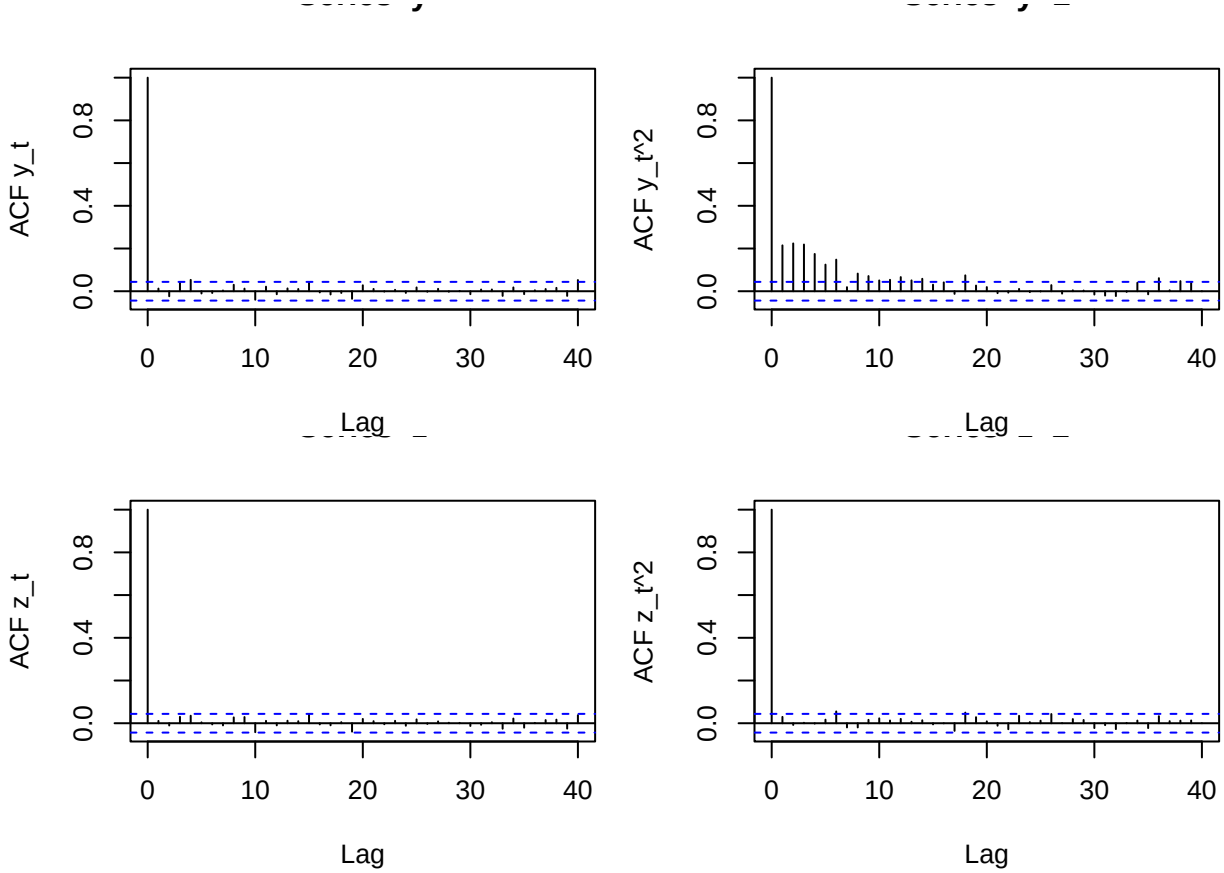

```r
plot(sig2, type="l", main="Conditional variance sigma_t^2")
```




```
par(mfrow=c(1,2), mar=c(4,4,2,1))
hist(y, breaks=40, main="Histogram: original series y_t", xlab="y_t")
hist(z, breaks=40, main="Histogram: standardized series z_t", xlab="z_t")
```




```
par(mfrow=c(2,2))
acf(y,    lag.max=40, ylab="ACF y_t")
acf(y^2,  lag.max=40, ylab="ACF y_t^2")
acf(z,    lag.max=40, ylab="ACF z_t")
acf(z^2,  lag.max=40, ylab="ACF z_t^2")
```



The standardized series follows a normal distribution better than the original series. The ACF of y_t^2 is relevant and shows that a GARCH model is appropriate. The non-squared ACFs are not relevant, which shows why squaring y_t is necessary.

Problem 6

```
set.seed(144)
spec_low <- ugarchspec(
  variance.model = list(model="sGARCH", garchOrder=c(1,1)),
  mean.model = list(armaOrder=c(0,0), include.mean=TRUE),
  distribution.model="norm"
)

setfixed(spec_low) <- list(
  mu = 0,
  omega = 0.1,
  alpha1 = 0.15,
  beta1  = 0.25
)

sim_low <- ugarchpath(spec_low, n.sim=2000, n.start=500)

y_low  <- as.numeric(fitted(sim_low)) + as.numeric(sim_low@path$seriesSim)
sig2_low <- as.numeric(sigma(sim_low))^2

z_low <- y_low / sqrt(sig2_low)
```

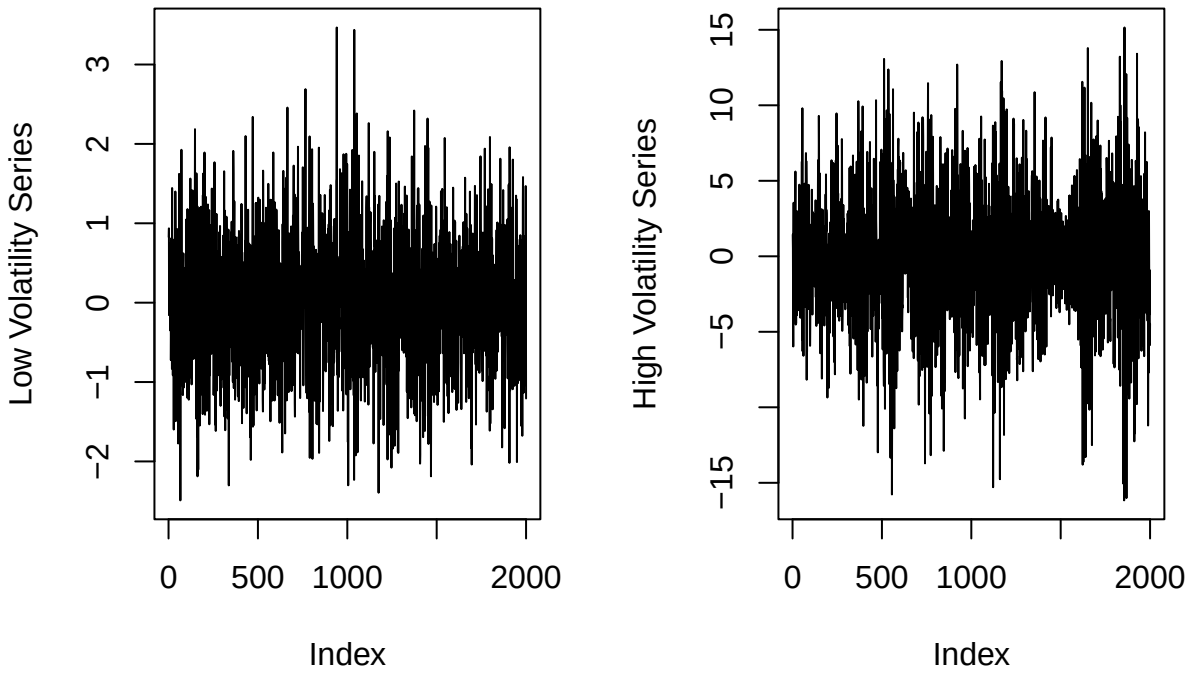
```r
spec_high <- ugarchspec(
  variance.model = list(model="sGARCH", garchOrder=c(1,1)),
  mean.model = list(armaOrder=c(0,0), include.mean=TRUE),
  distribution.model="norm"
)

setfixed(spec_high) <- list(
  mu = 0,
  omega = 0.1,
  alpha1 = 0.10,
  beta1  = 0.88
)

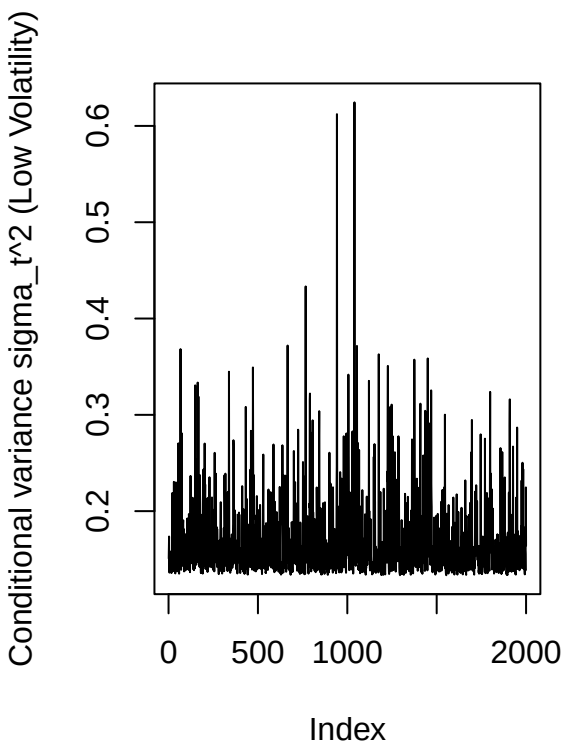
sim_high <- ugarchpath(spec_high, n.sim=2000, n.start=500)

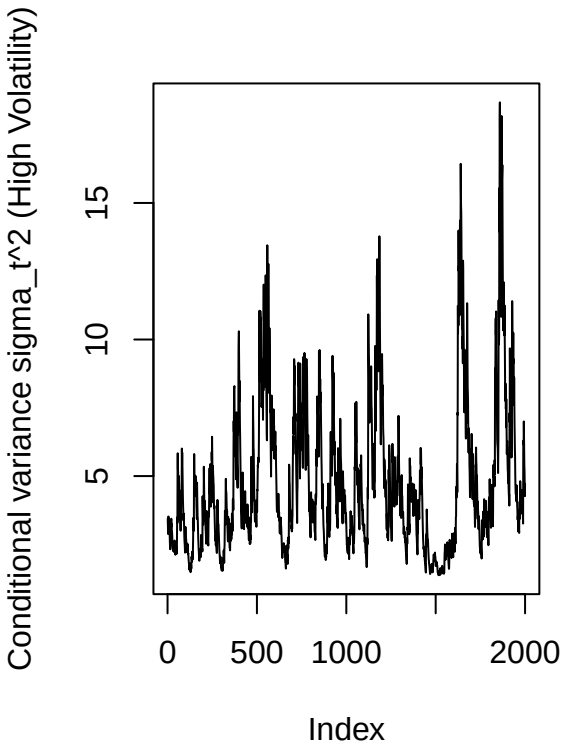
y_high  <- as.numeric(fitted(sim_high)) + as.numeric(sim_high@path$seriesSim)
sig2_high <- as.numeric(sigma(sim_high))^2
z_high <- y_high / sqrt(sig2_high)

par(mfrow=c(1,2))
plot(y_low, type="l", ylab="Low Volatility Series")
plot(y_high, type="l", ylab="High Volatility Series")
```

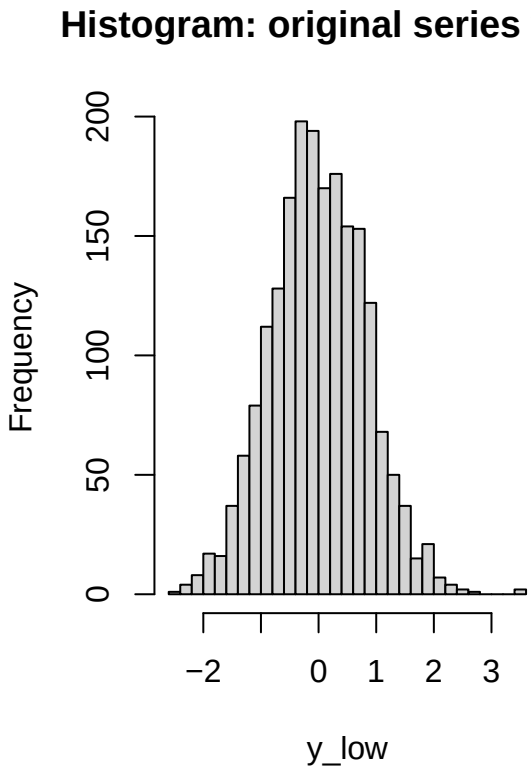


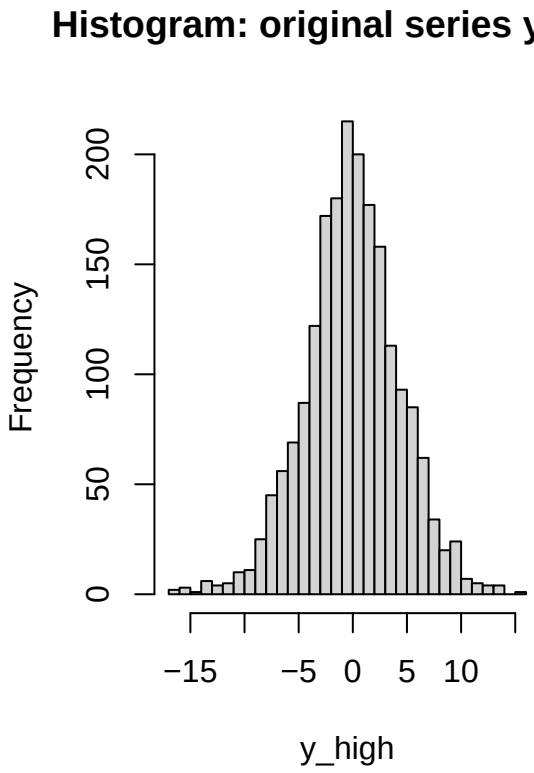

```r
plot(sig2_low, type="l", ylab="Conditional variance sigma_t^2 (Low Volatility)")
plot(sig2_high, type="l", ylab="Conditional variance sigma_t^2 (High Volatility)")
```



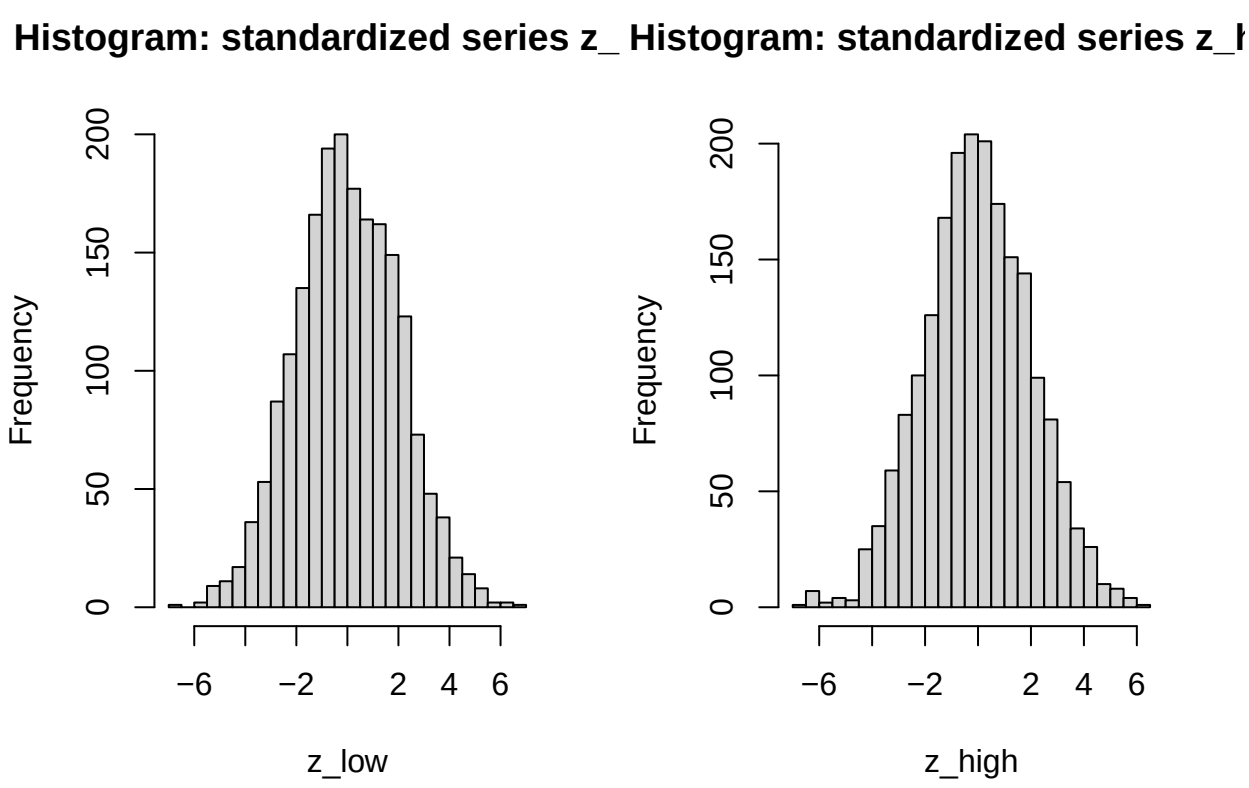



```r
hist(y_low, breaks=40, main="Histogram: original series y_low")
hist(y_high, breaks=40, main="Histogram: original series y_high")
```

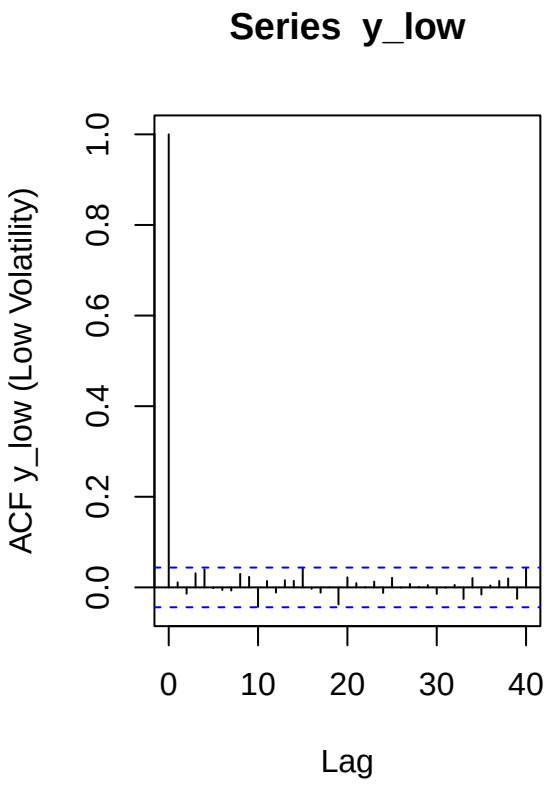


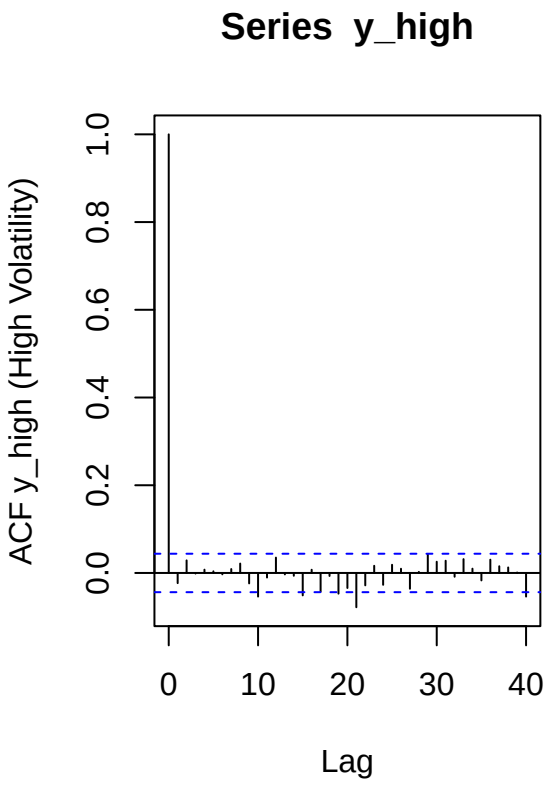



```
hist(z_low, breaks=40, main="Histogram: standardized series z_low")
hist(z_high, breaks=40, main="Histogram: standardized series z_high")
```

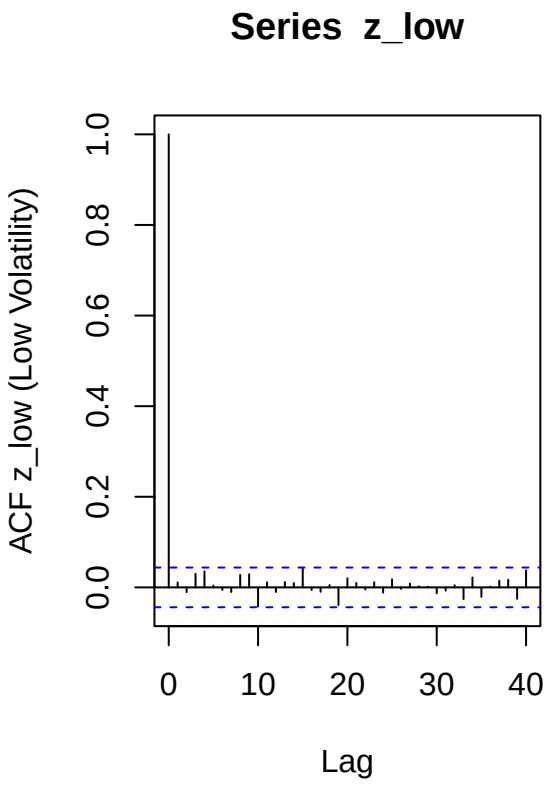


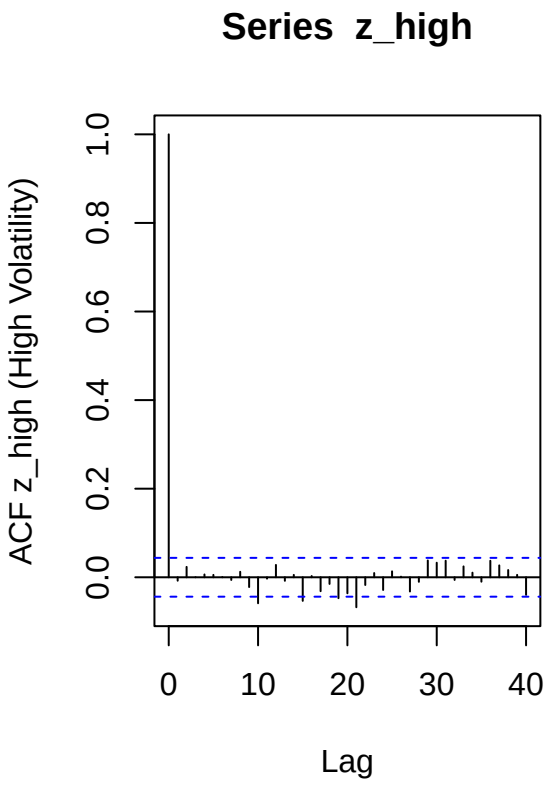

```
acf(y_low,    lag.max=40, ylab="ACF y_low (Low Volatility)")
acf(y_high,    lag.max=40, ylab="ACF y_high (High Volatility)")
```



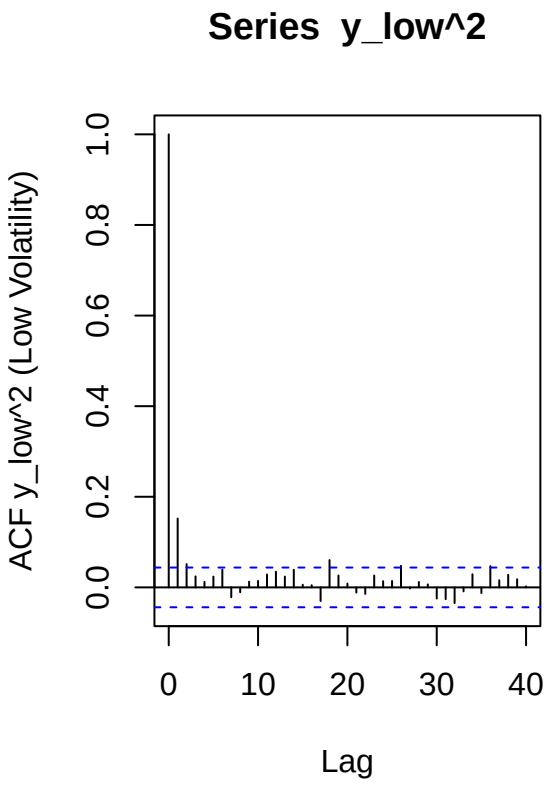


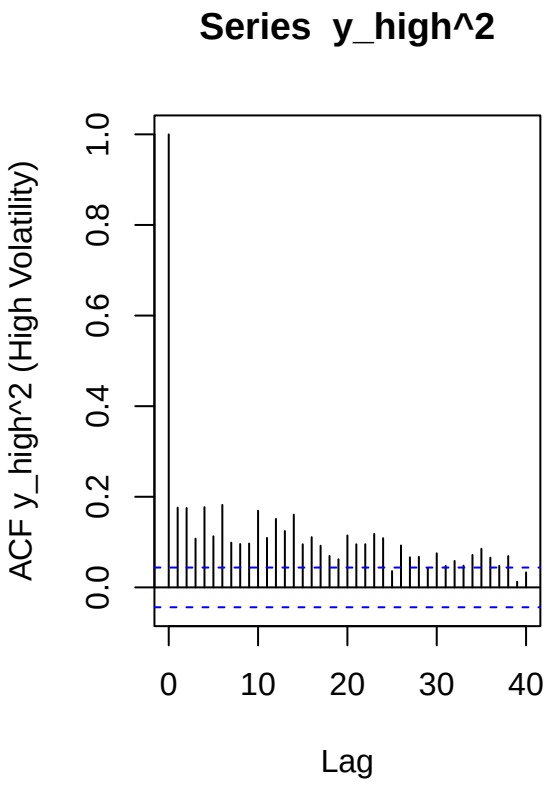

```r
acf(z_low,    lag.max=40, ylab="ACF z_low (Low Volatility)")
acf(z_high,    lag.max=40, ylab="ACF z_high (High Volatility)")
```



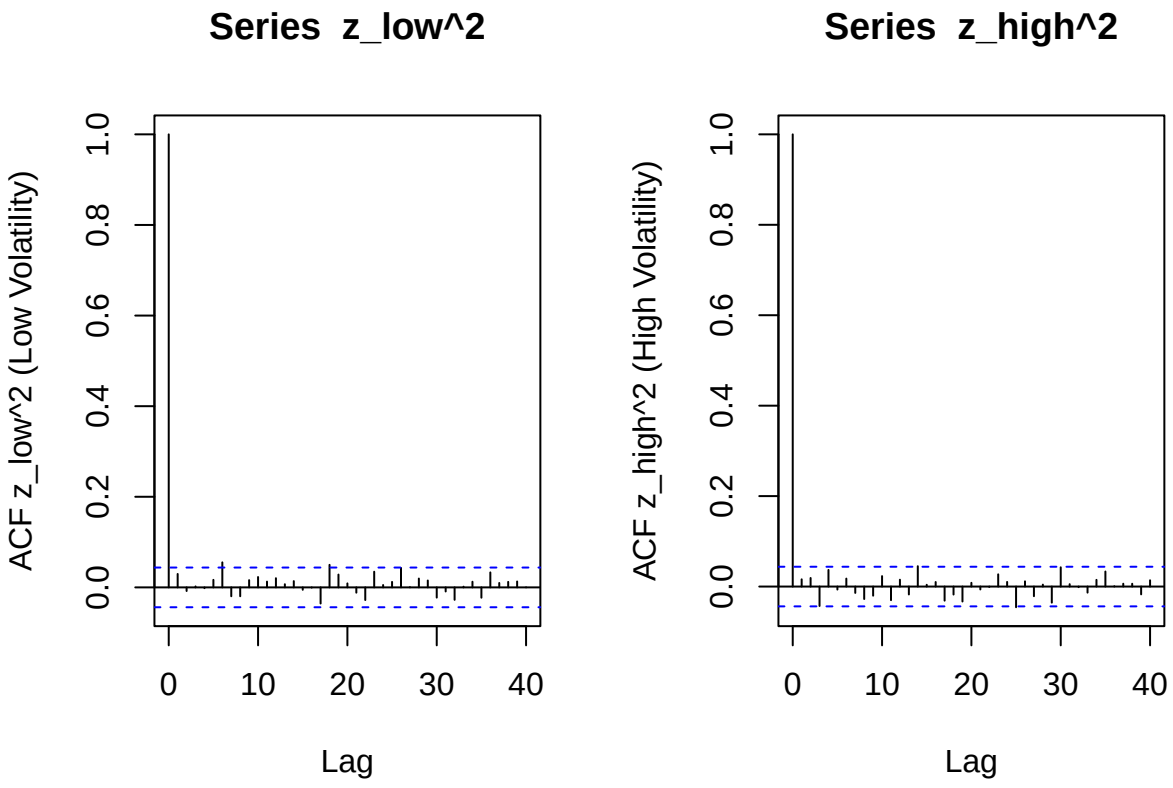



```r
acf(y_low^2,  lag.max=40, ylab="ACF y_low^2 (Low Volatility)")
acf(y_high^2,  lag.max=40, ylab="ACF y_high^2 (High Volatility)")
```






```r
acf(z_low^2,  lag.max=40, ylab="ACF z_low^2 (Low Volatility)")
acf(z_high^2,  lag.max=40, ylab="ACF z_high^2 (High Volatility)")
```



The two histograms are similar, however, the standardized series follows a normal distribution better than the original series. The ACF of the Yt^2 shows that there are volatility dynamics in the series and that a garch model is appropriate. This cannot be seen in the non-squared ACFs.

Problem 7

```
model=ugarchspec(
variance.model = list(model = "sGARCH", garchOrder = c(2, 2)),
mean.model = list(armaOrder = c(2, 2), include.mean = TRUE),
distribution.model = "norm")
getSymbols("^GSPC", src = "yahoo", from = "2000-01-01", to = "2025-12-31")
```

```
## [1] "GSPC"
```

```
sp500 <- na.omit(Ad(GSPC))
sp500_old <- sp500["2000-01-01/2010-12-31"]
sp500_new <- sp500["2015-01-01/2025-12-31"]
sp500old <- ts(as.numeric(sp500_old),
               start = c(2000, 1),
               frequency = 252)
sp500new <- ts(as.numeric(sp500_new),
               start = c(2015, 1),
               frequency = 252)
head(sp500old)
```

```
## Time Series:
```

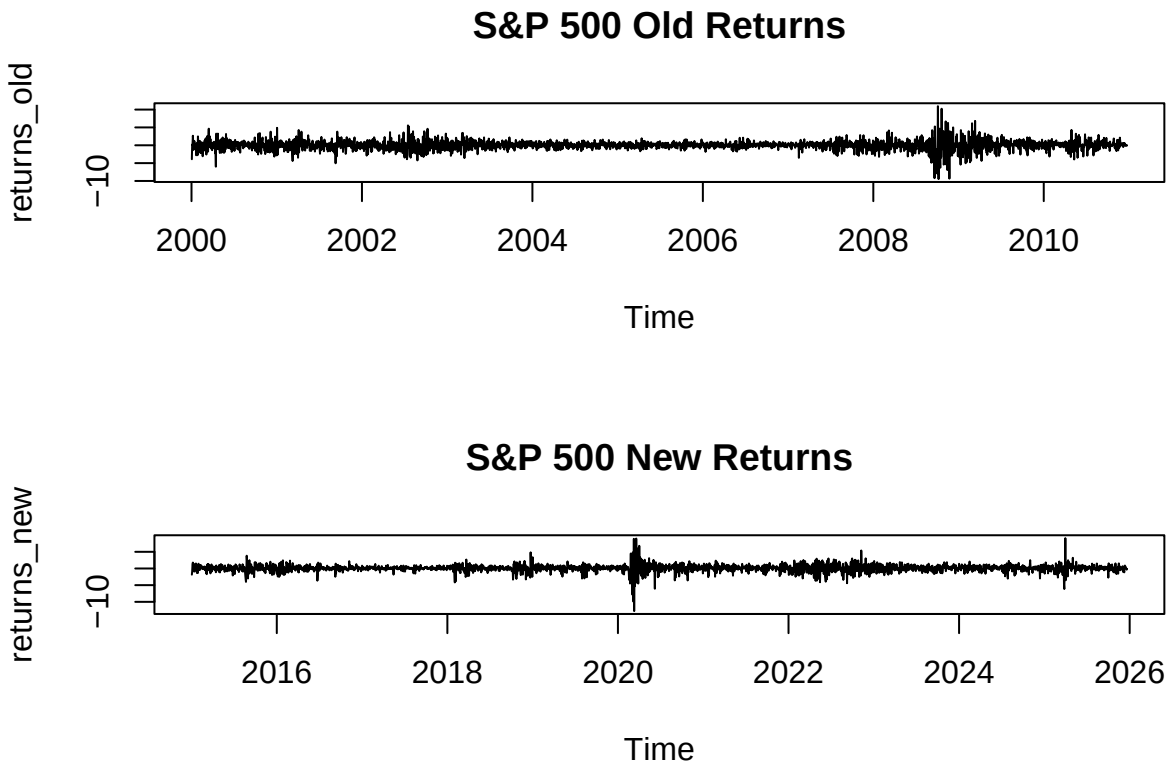
```
## Start = c(2000, 1)
## End = c(2000, 6)
## Frequency = 252
## [1] 1455.22 1399.42 1402.11 1403.45 1441.47 1457.60
```

```
head(sp500new)
```

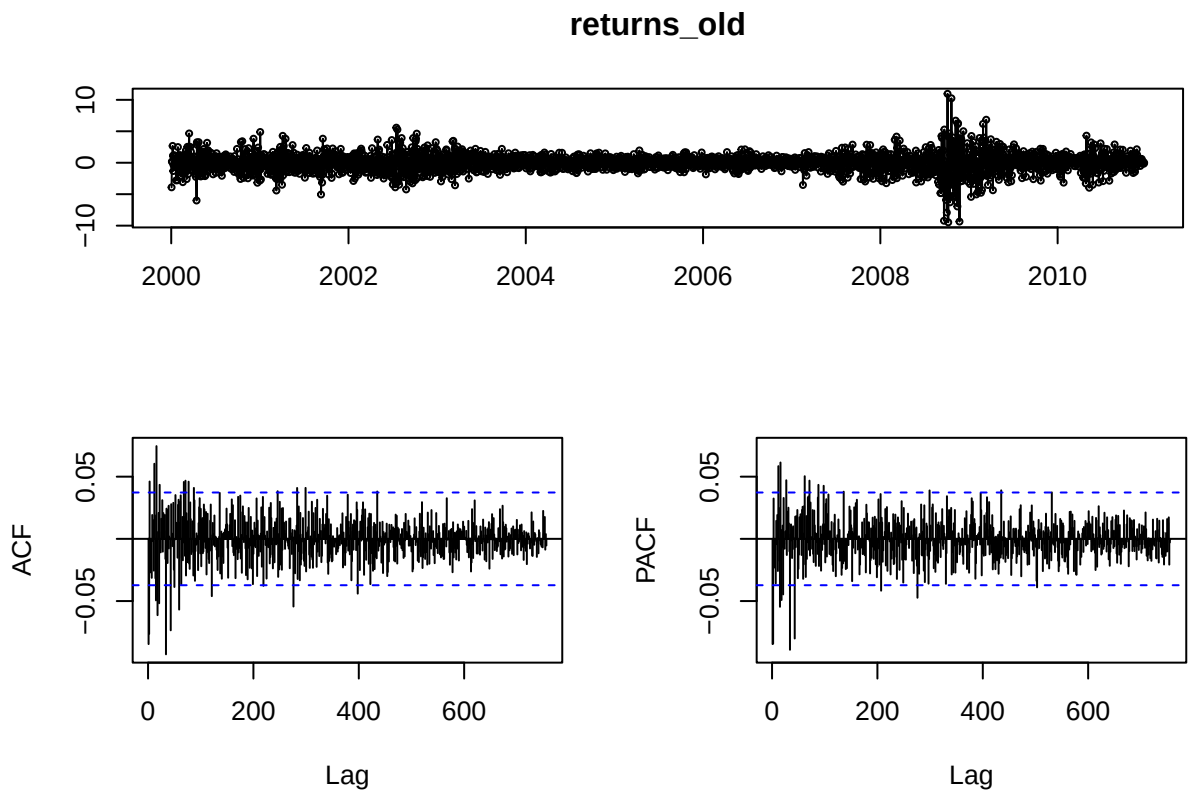
```
## Time Series:
## Start = c(2015, 1)
## End = c(2015, 6)
## Frequency = 252
## [1] 2058.20 2020.58 2002.61 2025.90 2062.14 2044.81
```

```
returns_old=diff(log(sp500old))*100
returns_new=diff(log(sp500new))*100

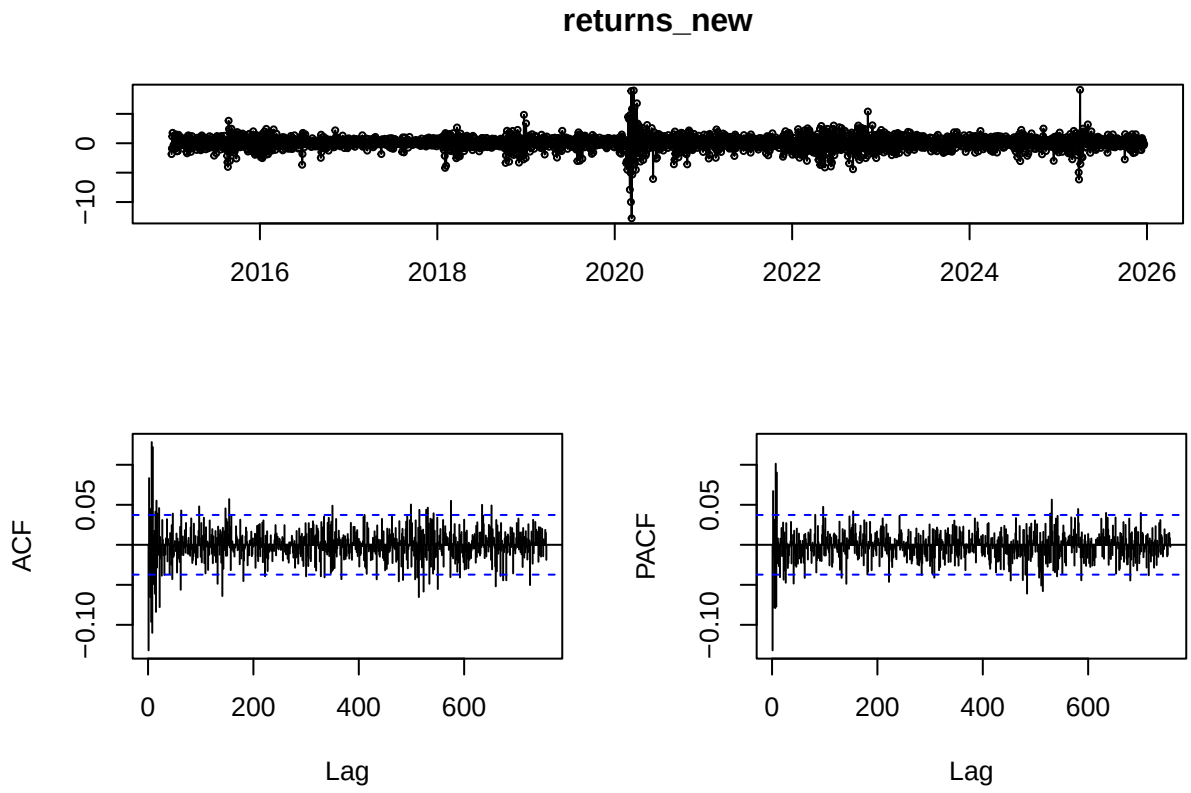
par(mfrow=c(2,1))
plot(returns_old, main="S&P 500 Old Returns")
plot(returns_new, main="S&P 500 New Returns")
```



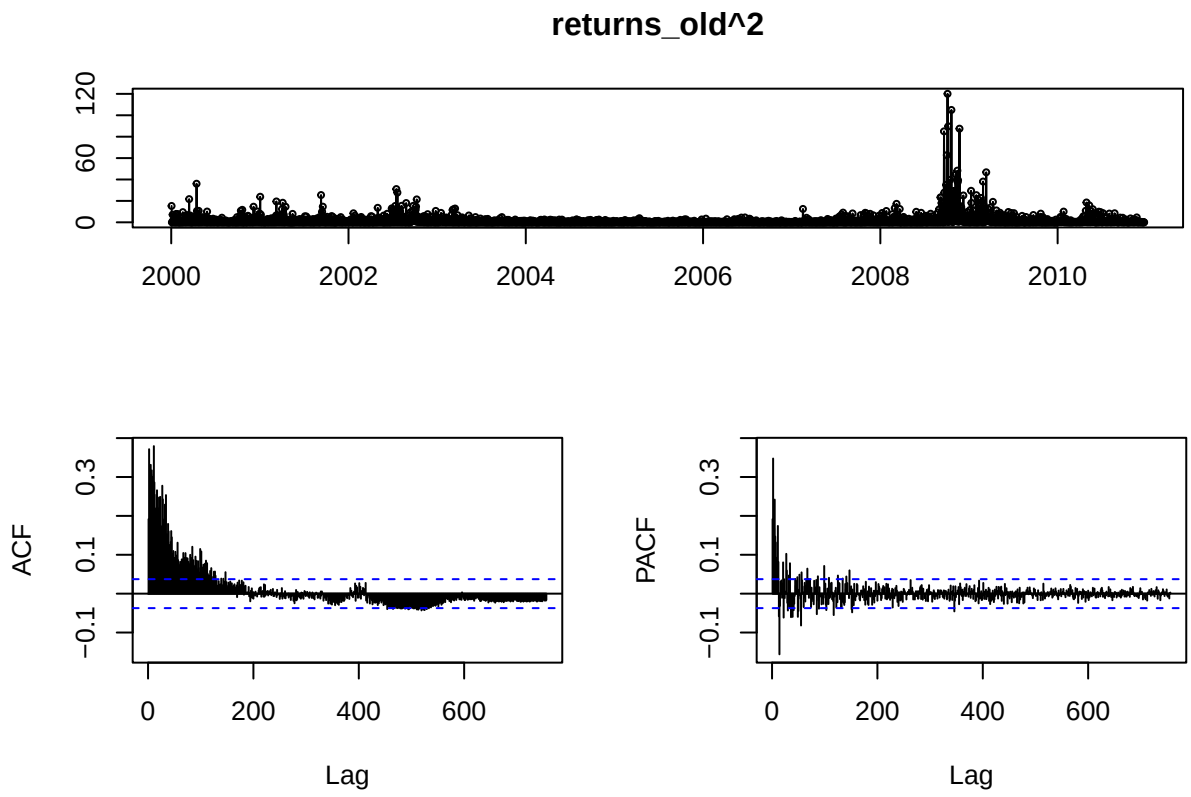

```
tsdisplay(returns_old)
```



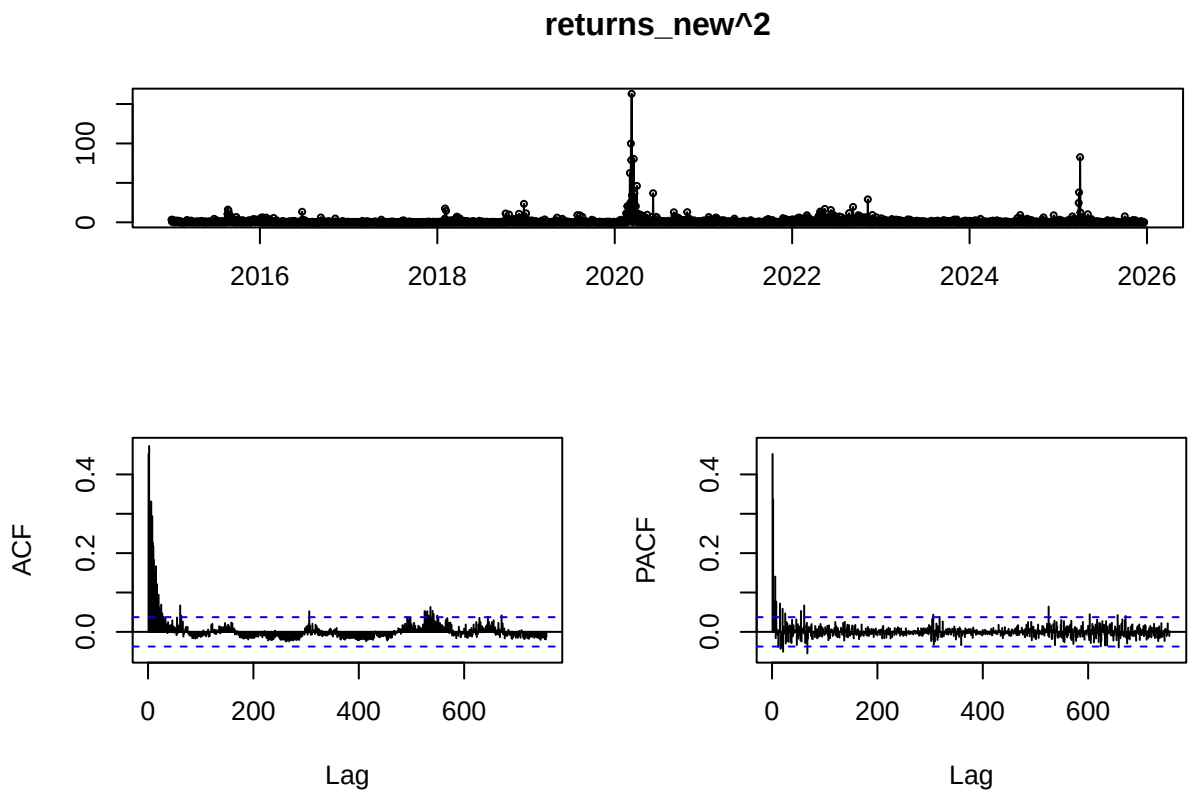

```
tsdisplay(returns_new)
```




```
tsdisplay(returns_old^2)
```




```
tsdisplay(returns_new^2)
```




```
modelfit=ugarchfit(spec=model,data=returns_new)
modelfit
```

```
##
## *---------------------------------*
## *          GARCH Model Fit        *
## *---------------------------------*
##
## Conditional Variance Dynamics
## -----------------------------------
## GARCH Model  : sGARCH(2,2)
## Mean Model   : ARFIMA(2,0,2)
## Distribution : norm
##
## Optimal Parameters
## ------------------------------------
##         Estimate  Std. Error     t value Pr(>|t|)
## mu      0.076965    0.007522  1.0232e+01 0.000000
## ar1     0.372707    0.017472  2.1331e+01 0.000000
## ar2     0.560639    0.017241  3.2519e+01 0.000000
## ma1    -0.429874    0.000068 -6.2867e+03 0.000000
## ma2    -0.535440    0.000016 -3.4077e+04 0.000000
## omega   0.046671    0.017828  2.6178e+00 0.008849
## alpha1  0.186991    0.026739  6.9933e+00 0.000000
## alpha2  0.024544    0.079497  3.0874e-01 0.757520
## beta1   0.543653    0.412881  1.3167e+00 0.187929
```

```
## beta2   0.207189    0.327067  6.3347e-01 0.526424
##
## Robust Standard Errors:
##         Estimate  Std. Error    t value Pr(>|t|)
## mu      0.076965    0.007874     9.7746 0.000000
## ar1     0.372707    0.017181    21.6931 0.000000
## ar2     0.560639    0.016607    33.7587 0.000000
## ma1    -0.429874    0.000088 -4912.5927 0.000000
## ma2    -0.535440    0.000083 -6428.0806 0.000000
## omega   0.046671    0.014684     3.1782 0.001482
## alpha1  0.186991    0.042996     4.3490 0.000014
## alpha2  0.024544    0.050039     0.4905 0.623782
## beta1   0.543653    0.164086     3.3132 0.000922
## beta2   0.207189    0.121705     1.7024 0.088683
##
## LogLikelihood : -3560.959
##
## Information Criteria
## ------------------------------------
##
## Akaike       2.5839
## Bayes        2.6053
## Shibata      2.5839
## Hannan-Quinn 2.5916
##
## Weighted Ljung-Box Test on Standardized Residuals
## ------------------------------------
##                          statistic p-value
## Lag[1]                       1.878  0.1706
## Lag[2*(p+q)+(p+q)-1][11]     4.148  0.9997
## Lag[4*(p+q)+(p+q)-1][19]     7.063  0.9026
## d.o.f=4
## H0 : No serial correlation
##
## Weighted Ljung-Box Test on Standardized Squared Residuals
## ------------------------------------
##                          statistic p-value
## Lag[1]                      0.5428  0.4613
## Lag[2*(p+q)+(p+q)-1][11]    5.9200  0.4468
## Lag[4*(p+q)+(p+q)-1][19]    8.9910  0.5627
## d.o.f=4
##
## Weighted ARCH LM Tests
## ------------------------------------
##             Statistic Shape Scale P-Value
## ARCH Lag[5]     0.612 0.500 2.000  0.4340
## ARCH Lag[7]     1.516 1.473 1.746  0.6187
## ARCH Lag[9]     2.736 2.402 1.619  0.6150
##
## Nyblom stability test
## ------------------------------------
## Joint Statistic:  2.4721
## Individual Statistics:
## mu     0.40550
```

```
## ar1    0.02351
## ar2    0.01670
## ma1    0.03314
## ma2    0.04688
## omega  0.46440
## alpha1 0.22548
## alpha2 0.40018
## beta1  0.59033
## beta2  0.61208
##
## Asymptotic Critical Values (10% 5% 1%)
## Joint Statistic:          2.29 2.54 3.05
## Individual Statistic:     0.35 0.47 0.75
##
## Sign Bias Test
## -----------------------------------
##                    t-value      prob sig
## Sign Bias           3.3813 7.315e-04 ***
## Negative Sign Bias  0.1052 9.162e-01
## Positive Sign Bias  0.3330 7.391e-01
## Joint Effect       21.9315 6.741e-05 ***
##
##
## Adjusted Pearson Goodness-of-Fit Test:
## ------------------------------------
##   group statistic p-value(g-1)
## 1    20     151.1    1.339e-22
## 2    30     164.8    6.257e-21
## 3    40     181.7    2.677e-20
## 4    50     192.6    6.435e-19
##
##
## Elapsed time : 3.22527
```

Returns from 2000-2010 has higher volatility in returns compared to returns from 2015-2025. There are definitely times of high and low volatility throughout both periods.

The model after the PACF of the Yˆ2 looks like it could be a very high order ARCH model. After running ugarch, a garch(2,2) can also be used in conjunction with an arma(2,2) model.

Problem 8

```
modelfor = ugarchforecast(modelfit, data = NULL, n.ahead = 2, n.roll =0 , out.sample = 0)
modelfor
```

```
##
## *----------------------------------*
## *       GARCH Model Forecast       *
## *----------------------------------*
## Model: sGARCH
## Horizon: 2
## Roll Steps: 0
## Out of Sample: 0
##
```

```
## 0-roll forecast [T0=1977-07-27]:
##       Series  Sigma
## T+1 0.08221 0.6153
## T+2 0.07930 0.6414
```

```
plot(modelfor, which=1)
```

